# 4. User manual

This chapter explains, for users, how to install QCalc 1.4.0 and how to calculate, use calculus (derivatives, integrals, limits), matrices, graphs, the menu and the settings. The app itself also has examples that you can tap to try: "Examples and tips" in the menu.

The screenshots in this chapter show version 1.4.0 in English (the screens that 1.4.0 did not change were taken with 1.3.0). They were taken on a PC: Chromium showed the app's own screen files (index.html, app.js, style.css) at the size of a phone (412 × 854), and the calculations ran on the same arm64 calculation engine as on a phone, under qemu (the screenshots of the suggestions and of the very long result used the same engine built for the PC). They were not taken on an AQUOS sense8, so text sizes differ slightly from a phone, and Android's status bar and navigation bar are not shown. The automatic update of exchange rates was off.

## 4.1 Installing and updating

### What you need

- A 64-bit (arm64) phone with Android 7.0 or later. QCalc is made for the SHARP AQUOS sense8 (Android 16).
- Where to get it: Google Play (a download of about 3.4 MB)

### Installing

1. Search for "QCalc" in the Google Play Store, or open the link you were given. Until the app is published, testers get it from the test link with the Google account that is registered as a tester.
2. Tap "Install".
3. Open "QCalc" from the app list. For a few seconds after it starts, "Starting the calculation engine…" is shown. When the phone is online, the exchange rates are updated automatically.

### Language

QCalc starts in English. To use it in Japanese, open the menu (☰) and choose "🌐 Language / 言語", or choose the language at the top of the settings. The app opens again in that language; the history and the expression being typed stay.

### Updating

Update from Google Play (automatically when auto-update is on, otherwise with "Update" in the store). The history, the settings, your definitions and ans are kept. The developer's test build (named "QCalc (debug)") installs as a separate app with its own data.

### Uninstalling

Settings app → "Apps" → "QCalc" → "Uninstall" (or from the app's page in Google Play). The history, the settings, your definitions and the downloaded exchange rates are all removed.

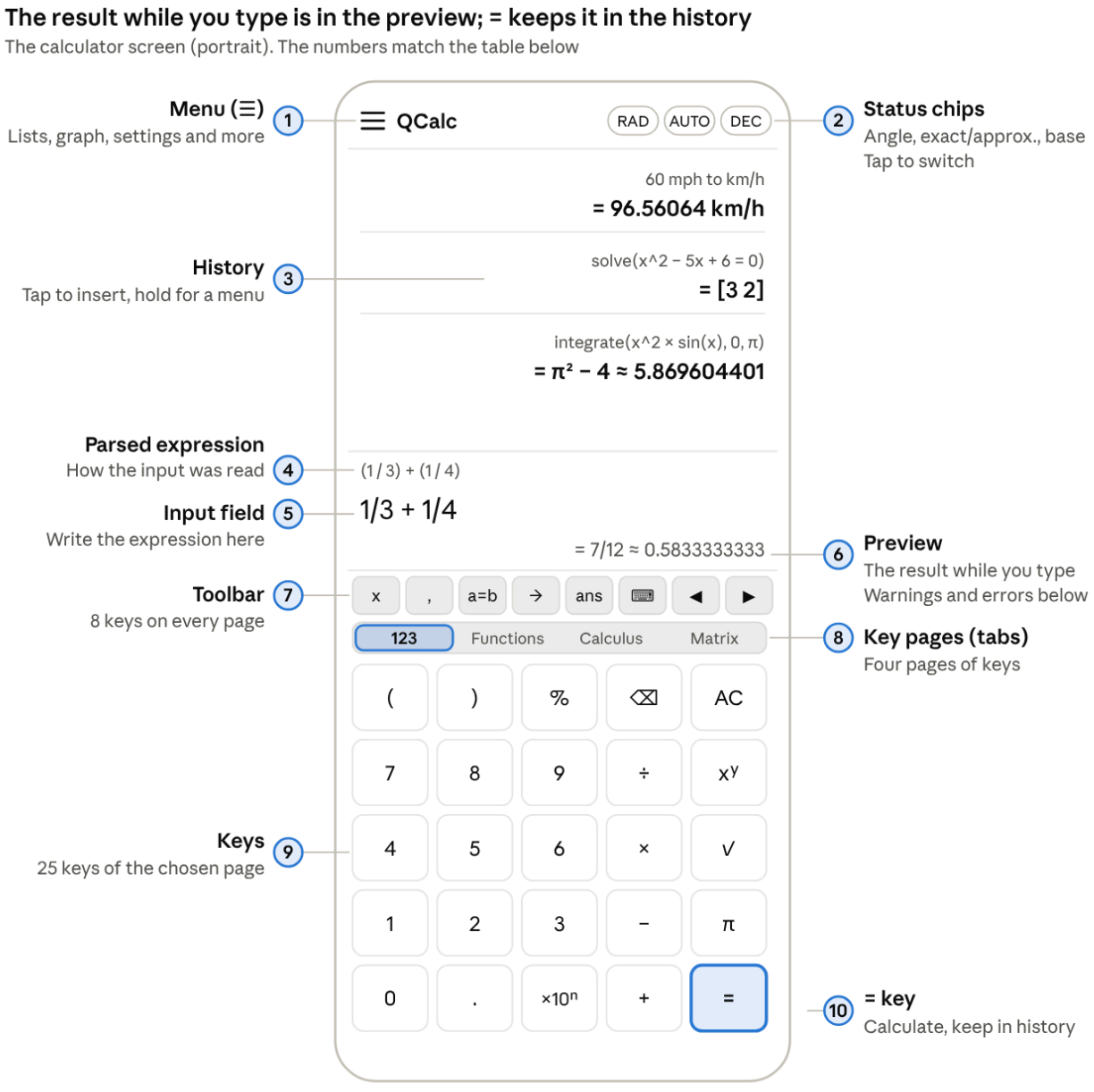
### Permissions

| Permission | Used for |
|---|---|
| Internet | Only downloading exchange rates. Expressions and the history are never sent |
| Vibration | A short vibration when a key is pressed (can be turned off in the settings) |

QCalc does not use contacts, storage, location, the camera or other permissions. To export the history, Android's own "save" screen is used, so no storage permission is needed.

QCalc collects no personal data and has no ads. See the privacy policy (https://hiroshitsu.github.io/qcalc/privacy.html, also opened from "About this app" in the app).

## 4.2 The screen

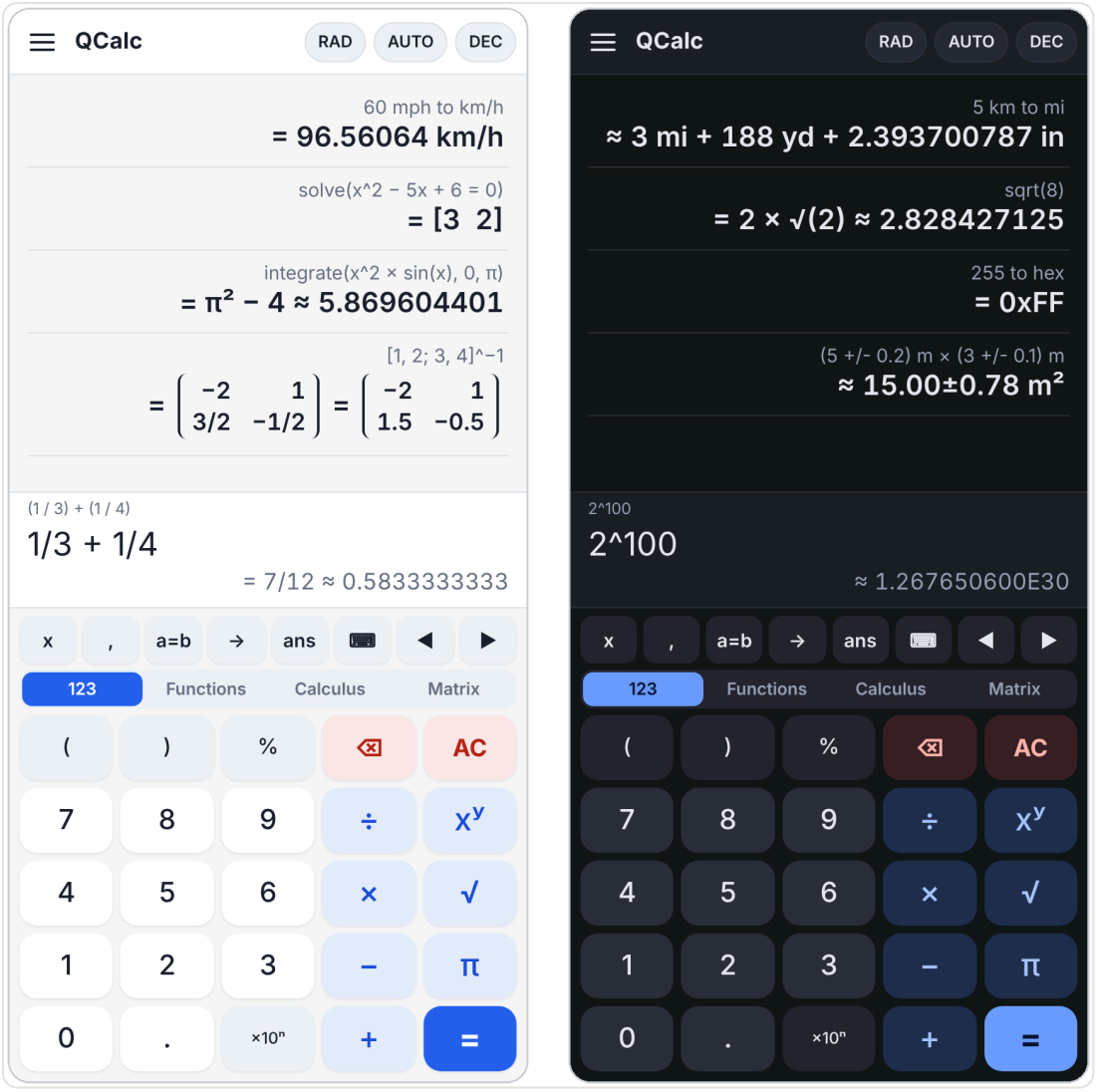



The calculator screen · the names and jobs of its 10 parts

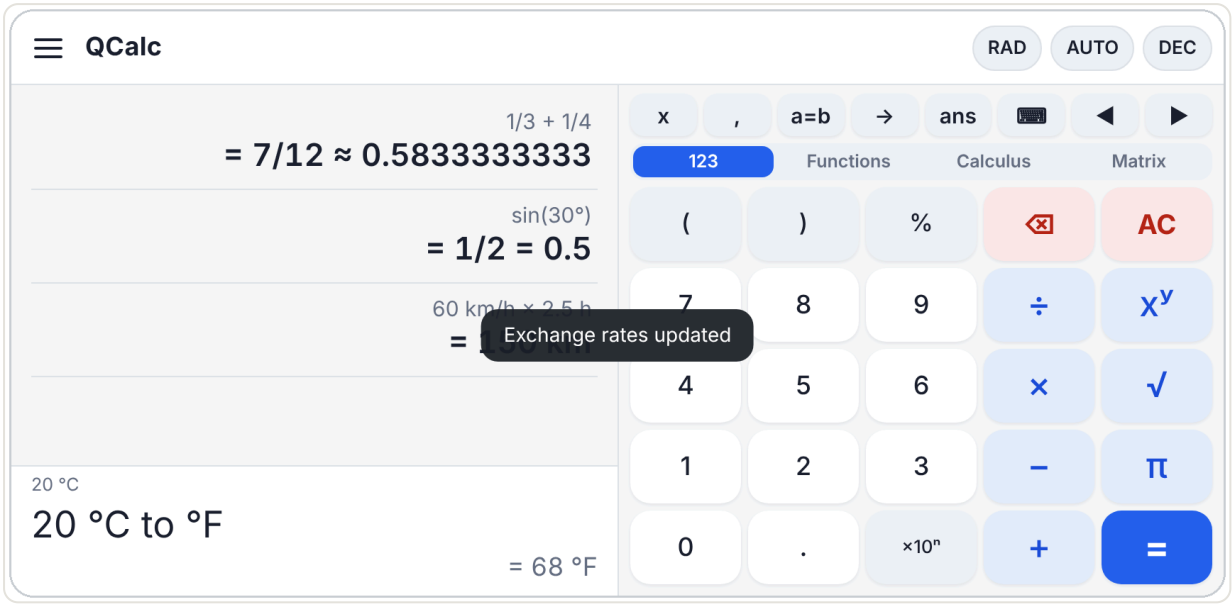
| No. | Name | Description |
|---|---|---|
| 1 | Menu (☰) | Opens the lists of functions, units and variables, the graph, user definitions, the exchange-rate update, the settings and more (4.8) |
| 2 | Status chips | From the left: the angle unit (RAD, DEG, GRA), how to calculate (AUTO, EXACT, APPROX) and the number base of results (DEC, HEX, BIN, OCT). Each tap switches to the next. Duodecimal and Roman numerals are chosen in the settings |
| 3 | History | The expressions calculated with = and their results; new ones are added at the bottom (up to 300). Tap an expression to edit it again, tap a result to insert it, touch and hold for a menu (copy, delete and more). Matrix results are shown as tables. Before anything is calculated, it shows how to start and examples that you can tap |
| 4 | Parsed expression | How the engine read the input, with parentheses and the full names of units. Use it to check that the input was read as you meant. Right after AC, or after an expression was replaced from the history or the examples, "Undo" appears at the right of this line (4.3) |
| 5 | Input field | Where you write the expression. While you type with the on-screen keys, the phone's keyboard stays hidden (⌨ switches, 4.6). The expression being typed is kept when the app closes |
| 6 | Preview and messages | The result while you type (before =). Matrices are shown as tables. Warnings and errors appear below it (errors in red). After →, and while you type a name with the keyboard, suggestions that a tap types appear below them (4.3, 4.6). Definitions show "Press = to run it (definitions are not previewed)". During a long calculation, tap here to stop it |
| 7 | Toolbar | Keys that are always there: x, comma, a=b (the "=" of an equation), → ("to": convert), ans (the last result), ⌨ (the phone's keyboard), ◀ ▶ (move the cursor) |
| 8 | Key pages (tabs) | 123, Functions, Calculus and Matrix. Tap one to change the keys below. The selected tab is colored |
| 9 | Keys | 25 keys of the selected page: numbers and operators on 123, trigonometric functions and logarithms on Functions, derivatives, integrals and limits on Calculus, entering matrices and their determinant and inverse on Matrix (the figure in 4.3, 4.4, 4.5) |
| 10 | = key | Calculates and keeps the result in the history. Enter on the phone's keyboard does the same |

- In landscape, the history and the input field are on the left and the keys on the right (when the screen is not tall).
- The colors follow the phone's dark theme. The setting "Light or dark" can keep the app always light or always dark (4.9). Seven color themes for the keys and highlights can be chosen in the settings (4.9).
- The back gesture closes what is open (a sheet → the menu → a list or another panel → the keyboard mode), one at a time. When nothing is open, the app moves to the background (the expression stays).

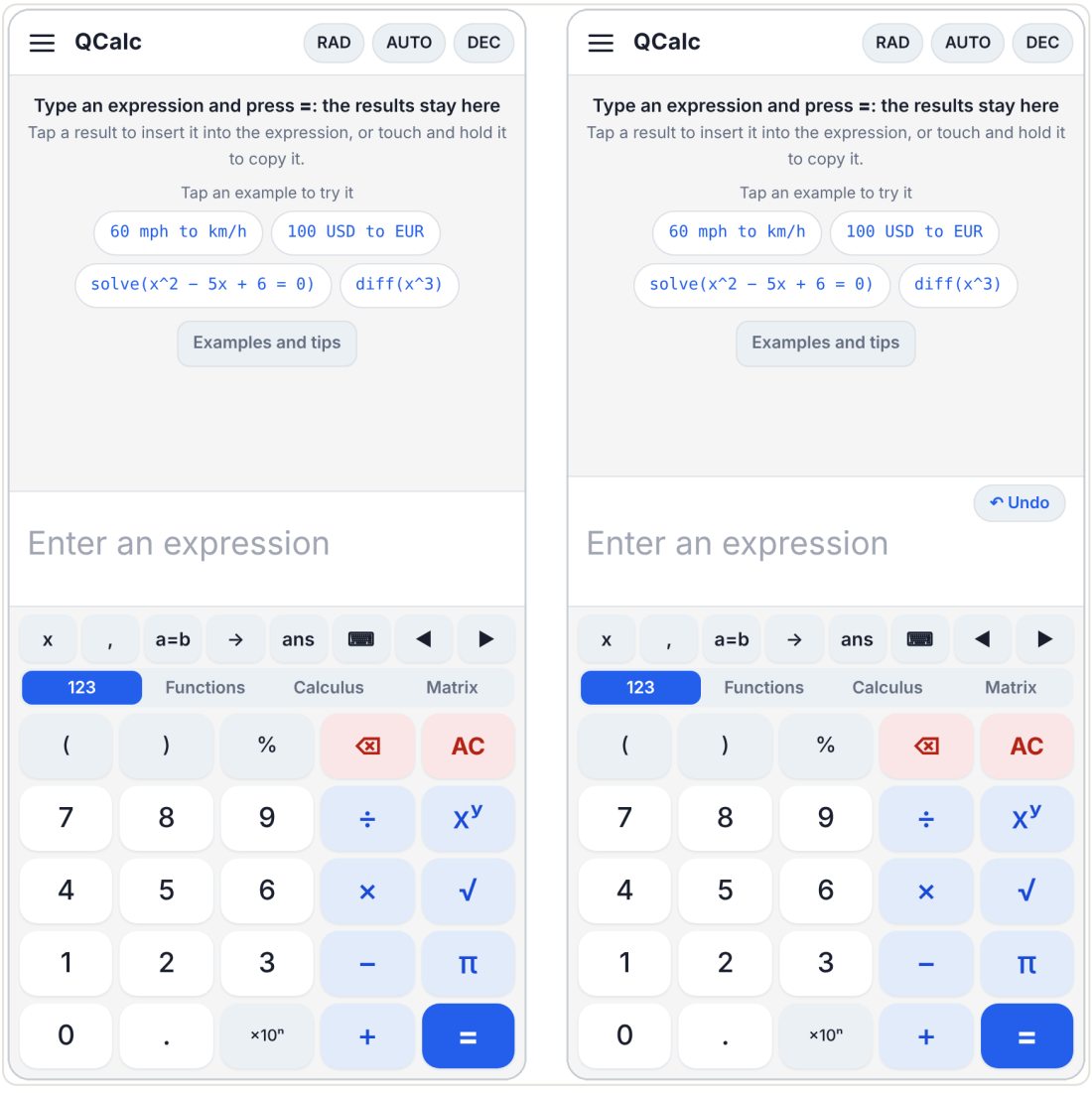
**Screen example** The calculator screen (left) and the same screen in the dark theme (right). On the left, "1/3 + 1/4" is being typed and its result is shown under the input field. The last history item is a matrix calculation, shown as a table.



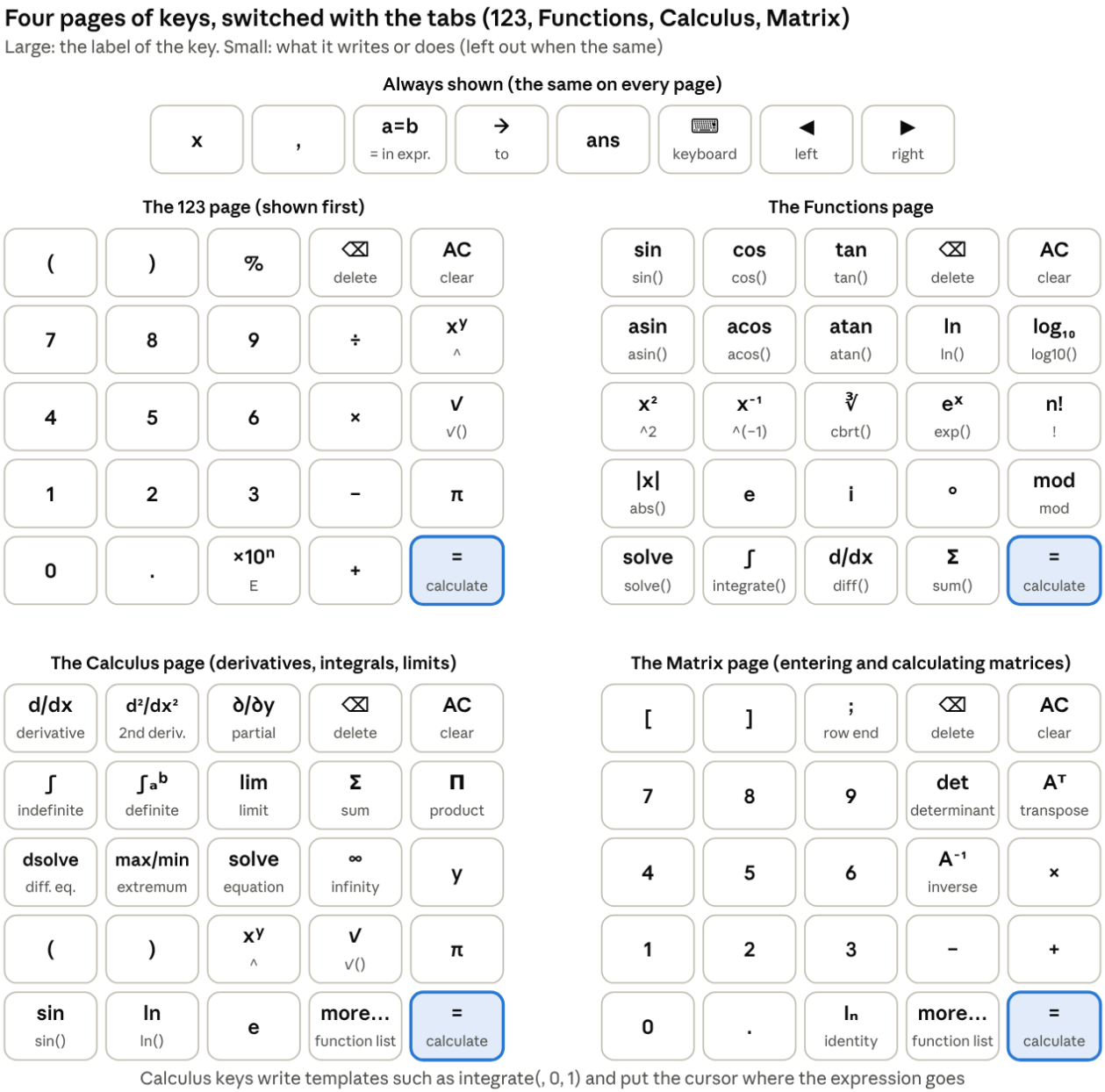
**Screen example** The calculator in landscape (the history and the input field on the left, the keys on the right).




**Screen example** The first start (left: how to start, and examples that you can tap) and right after AC cleared the expression (right: "Undo" at the top right of the input field).



# 4.3 Basic calculations




The keys · four pages: 123, Functions, Calculus, Matrix

There are four pages of keys, switched with the tabs right above the keys (123, Functions, Calculus, Matrix). The row above them (x, comma, a=b, →, ans, ⌨, ◀ ▶) works with every page. The Calculus keys are explained in 4.4 and the Matrix keys in 4.5.

### Calculating

1. Type the expression with the keys. While you type, the result is shown under the input field (the preview).
2. Press = to calculate. The expression and the result are kept in the history, and the input field is cleared.
3. To go on from the result, press ÷, ×, +, $x^y$ or another operator in the empty input field: the last result (ans) is put in front (for example "× 2 =" after =). − does not add ans, so that you can type a negative number.

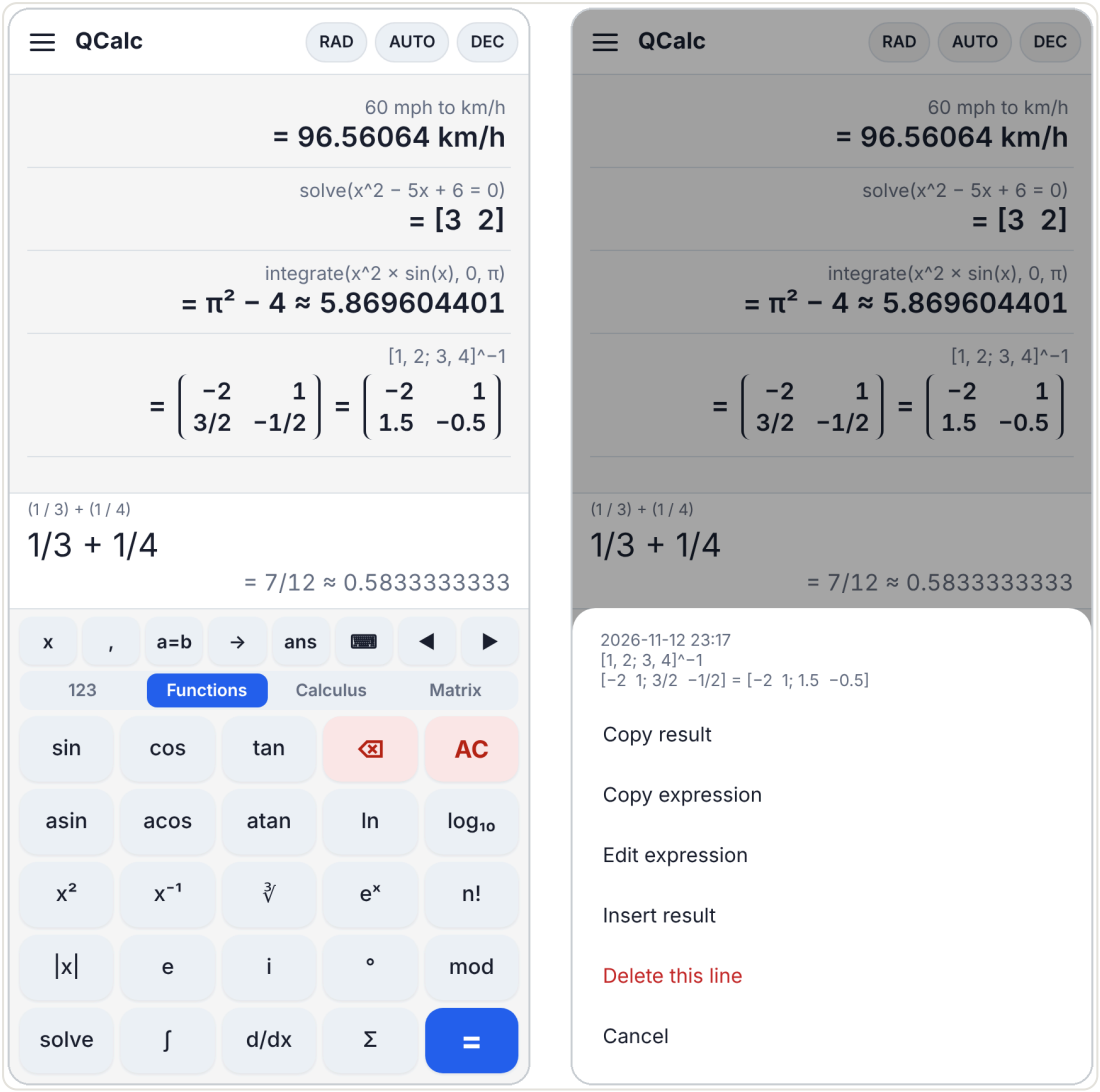
### Correcting the input

- ⌫ deletes the character before the cursor; hold it to keep deleting. A function name with its parenthesis, such as "sin(", is deleted at once when it is empty (so is a template such as "integrate(, 0, 1)" before anything is written in it).
- AC clears the input field (not the history). Right after it, "Undo" appears at the top right of the input field; tap it to get the expression back. It disappears when you type.
- ◀ ▶ move the cursor (hold to repeat). With text selected, they move to the start (◀) or the end (▶) of the selection. You can also tap the input field to place the cursor.
- A function key (such as sin) writes "sin()" and puts the cursor inside the parentheses. Some Calculus keys write a whole template, such as "integrate(, 0, 1)". With text selected, the key wraps the selection in the function. The a=b key writes the " = " of an equation (the big = key calculates). Example: solve(x^2 = 16)

### Using results

- ans is the last result; ans2 to ans5 are the results before it.
- Tap a result in the history to insert its value at the cursor. Even when the result shows "2 + 1/3 ≈ 2.333333333", the exact value (7/3) is inserted.
- Tap an expression in the history to put it back into the input field for editing.
- Touch and hold a history item for "Copy result", "Copy expression", "Edit expression", "Insert result" and "Delete this line". Above the menu you see the date and time of the calculation, the expression and the result. After a deletion, "Line deleted" and "Undo" appear at the bottom for 5 seconds (also after "Clear history" in the menu, for 6 seconds).

**Screen example** The Functions keys (left, chosen with the tab) and the menu of a history item (right).



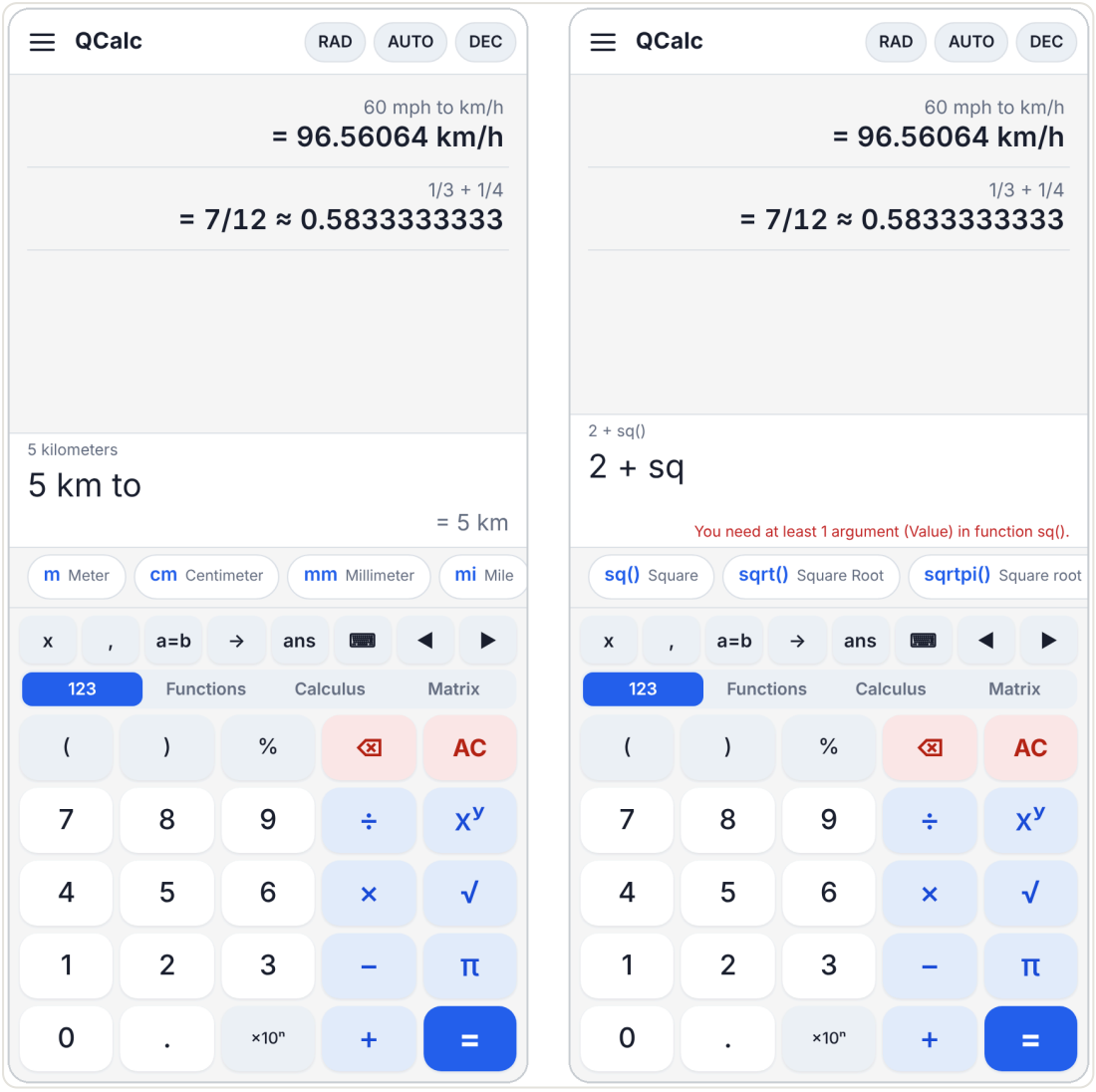

### Converting (→)

The → key writes " to ". Convert to units, currencies, number bases (hex and others), fractions (fraction) and more, as in "5 km → mi" (4.7).

When you press →, the units you can convert to are suggested below the input field (1.4.0). After "5 km →" you see m, cm, mm, mi, ft and more: the units of the same kind, the most used first. A tap writes the unit, and the result appears right below. The row scrolls sideways. If you do not need it, simply go on typing.

- The suggestions fit the unit in the expression: currencies for a currency, °F and K for a temperature, $m^2$, $km^2$, ha and acre for an area.
- For a number without a unit (255 →), the suggestions are ways to write a number: fraction, hex (hexadecimal), bin (binary), factors (prime factors) and others.
- Letters typed after → narrow the list to the units that begin with them (after "5 km → m": mm, mi and others).
- An expression with units of different kinds, such as "5 km / 2 h →", gets no suggestions. Type the target with the keys or take it from the unit list.
- While the expression ends with → alone, the preview shows the result of what is before it, and = calculates that.

Screen example  The conversion targets suggested after "5 km →" (left), and the names suggested after typing "2 + sq" with the keyboard (right; 4.6).




### Stopping a calculation

When a calculation takes time, "Calculating… (tap here to stop)" is shown; tap it to stop. A calculation started with = is stopped automatically after 20 seconds.

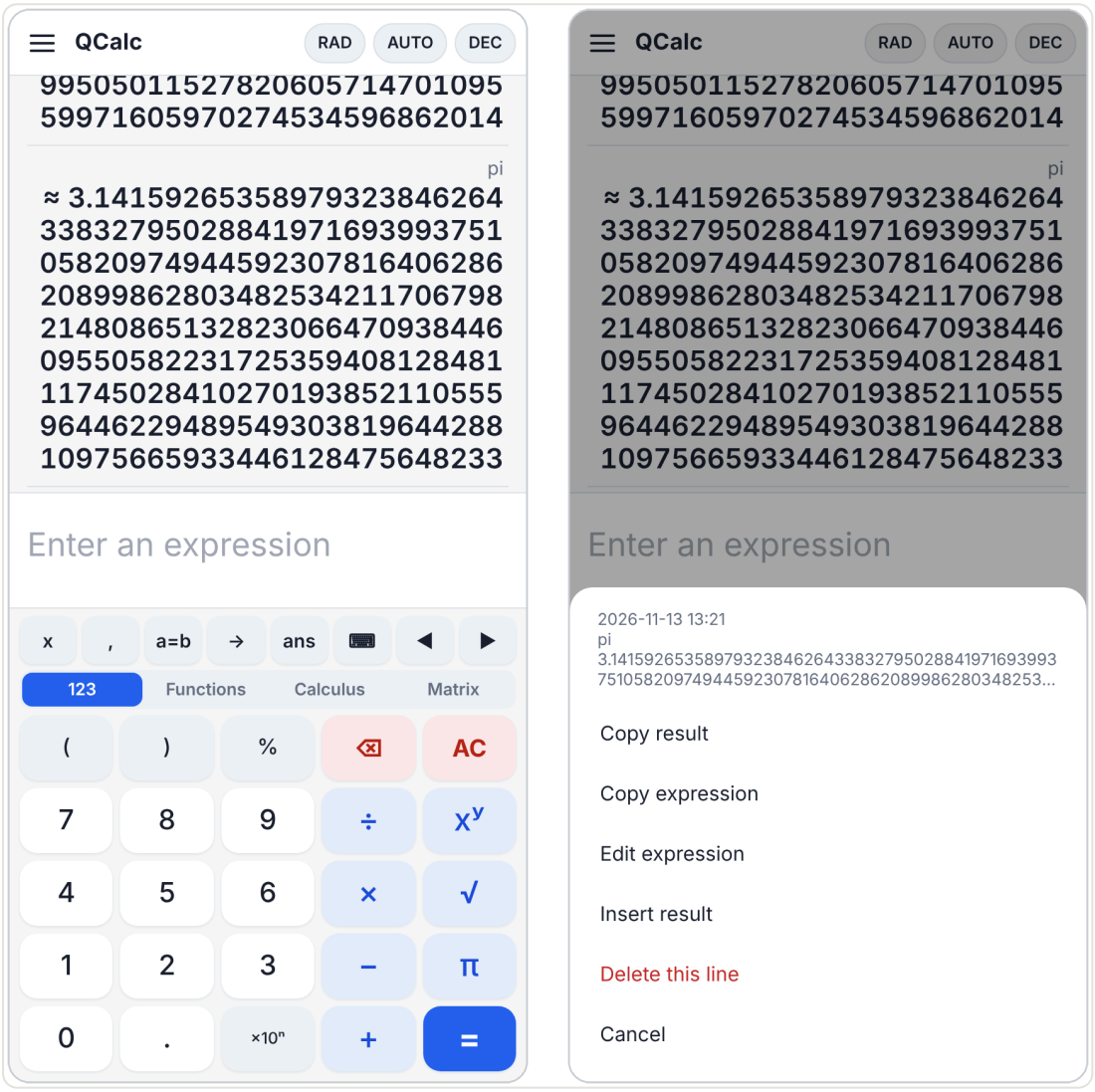
### Very long results (tens of thousands to 200,000 digits)

The precision (significant digits) in the settings goes up to 200,000. Use it to get π, $\sqrt{2}$ and the like with many digits.

- The preview while you type and the result in the history both show all the digits (up to about 450,000 characters). A long result sits in its own box in the history: move inside the box to read on; move outside it (an expression or another result) to scroll the whole history.
- "Copy result" in the menu of a history item copies all the digits, to paste into another app. Some very long results, such as a complex number with 200,000-digit parts, cannot be copied because of a limit in Android ("Too long to copy").
- A result longer than 20,000 characters is not inserted when tapped. Use the ans key instead; ans keeps all its digits even after the app is closed.

- A result longer than about 450,000 characters is shown as its beginning, "…" and its end, with the note "The middle of this long result is not shown (N digits)", both in the preview and in the history.
- The history keeps the newest results in full until they add up to 1,000,000 characters; older long results keep their first 2,000 characters and "…".
- More digits take longer. On a PC, 200,000 digits of π took 0.4 seconds and sin(1) 1.2 seconds; a phone can take longer. Heavy functions such as ζ(3) are stopped after 20 seconds.
- A high precision also slows down other calculations. Set it back when you are done (the default is 10).

**Screen example** π with a precision of 200,000 (left: the result scrolls in its box) and the menu of that result (right: "Copy result" copies all the digits).




### Status chips (top right)

- Angle: RAD → DEG → GRA. In DEG, sin(30) = 0.5.
- How to calculate: AUTO → EXACT → APPROX. EXACT keeps forms such as √(2); APPROX shows decimals.
- Number base: DEC → HEX → BIN → OCT. Results are shown in that base. In input, 0x (hexadecimal), 0b (binary) and 0o (octal) write a number in that base.

# 4.4 Calculus (derivatives, integrals, limits)

The Calculus page has keys for derivatives, integrals, limits, sums and differential equations. A key writes a template of the function and puts the cursor where the expression goes. Numbers are typed on the 123 page.

| Key | Writes | Meaning and example |
|---|---|---|
| d/dx | `diff()` | Derivative. `diff(x^3 + 2x)` = $3x^2 + 2$ |
| $d^2/dx^2$ | `diff(, x, 2)` | Second derivative (the last number is the order). `diff(x^5, x, 2)` = $20x^3$ |
| $\partial/\partial y$ | `diff(, y)` | Partial derivative with respect to y. `diff(x^2 × y^3, y)` = $3x^2y^2$ |
| $\int$ | `integrate()` | Indefinite integral. `integrate(1/(x^2 + 1))` = arctan(x) + C (C is the constant of integration) |
| $\int_a^b$ | `integrate(, 0, 1)` | Definite integral (lower bound, then upper bound). `integrate(x^2 × sin(x), 0, π)` = $\pi^2 - 4 \approx 5.869604401$ |
| lim | `limit(, 0)` | Limit (the second argument is the value approached). `limit(sin(x)/x, 0)` = 1 |
| Σ | `sum(, 1, 10)` | Sum. `sum(x^2, 1, 10)` = 385 |
| Π | `product(, 1, 10)` | Product. `product(x, 1, 5)` = 120 |
| dsolve | `dsolve(diff(y, x) = )` | First-order differential equation. `dsolve(diff(y, x) = −2y, 5)` solves $y' = -2y$ with y(0) = 5: 5 / e^(2x) |
| max/min | `extremum()` | The x of the extrema. `extremum(x^3 − 3x)` = [1 −1] |
| solve | `solve()` | Solves an equation. `solve(x^2 = 16)` = [4 −4] |
| ∞ | `∞` | Infinity. `integrate(e^(−x^2), −∞, ∞)` = $\sqrt{(\pi)} \approx 1.772453851$ |
| y | `y` | A second variable (for partial derivatives, double integrals and differential equations) |
| ( ) $x^y$ $\sqrt{}$ π sin ln e | — | Keys often needed in expressions (the same as on the 123 and Functions pages) |
| more… | — | Opens the list of functions at the category "Calculus" (numerical integration `romberg`, limits in several variables `multilimit`, functions defined by integrals such as Si and Ci) |

### Using the templates

- "0, 1" and "1, 10" in a template are placeholders. Move the cursor with ◀ ▶ and replace them.
- Select an expression and press a key to wrap it in the function (select `x^2` and press $\int_a^b$ → `integrate(x^2, 0, 1)`).
- A template with nothing written in it yet is deleted with one ⌫.
- The "=" of an equation is written with the a=b key in the row above.
- Use x, y and z as variables. Single letters such as t or m are read as units (tonne, metre).

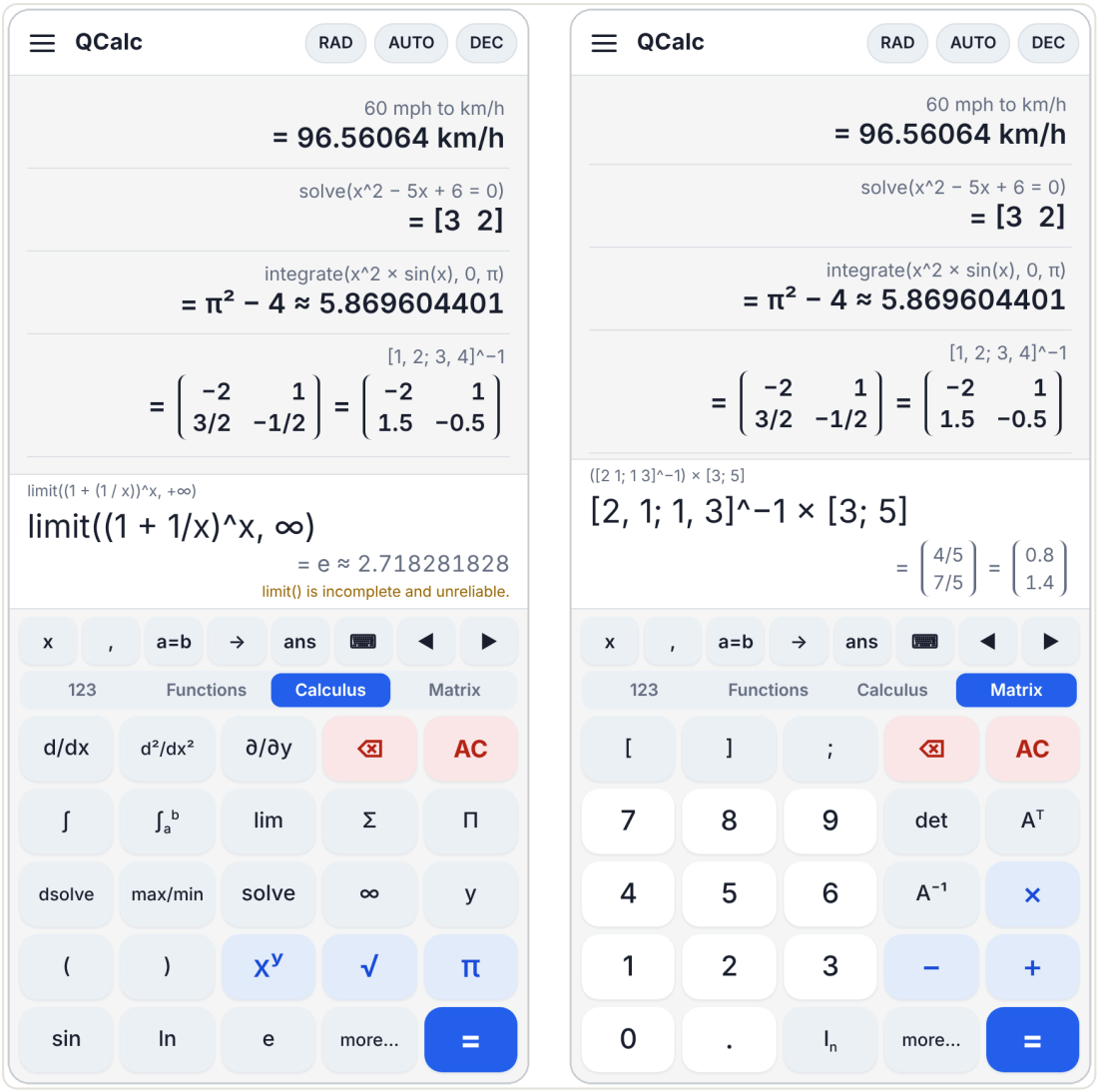
### More ways to use them

| To do this | Type | Result |
|---|---|---|
| Derivative at a point | `diff(x^3, x, 1, 2)` | 12 (the third argument is the order, the fourth the value of x) |
| Double integral | `integrate(integrate(x × y, 0, 1, x), 0, 2, y)` | 1 (the fourth argument is the variable) |
| Improper integral | `integrate(1/x^2, 1, ∞)` | 1 |
| Limit at infinity | `limit((1 + 1/x)^x, ∞)` | e ≈ 2.718281828 |
| One-sided limit | `limit(1/x, 0, x, 1)` | +∞ (the last 1 is from the right, −1 from the left) |
| Differential equation without an initial value | `dsolve(diff(y, x) = y)` | e^x × C (C is any constant) |
| Solve numerically | `newtonsolve(cos(x) = x, 1)` | ≈ 0.7390851332 (Newton's method, starting at 1) |
| Numerical integration | `romberg(sin(x)/x, 1, 2)` | ≈ 0.6593299064 |
| Special functions | `gamma(1/2)` , `zeta(2)` | √(π), π² / 6 |

### Good to know

- Every limit shows the note "limit() is incomplete and unreliable". The engine (libqalculate) always adds it to limits; it is not a fault.
- A definite integral without a closed form is calculated numerically, with the note "Definite integral was approximated" (for example `integrate(x^x, 0, 1)` ≈ 0.78343051). It takes time, so the result may only appear after =.
- An infinite sum (such as `sum(1/x^2, 1, ∞)` ) is returned as it is, without a value. Use a function such as `zeta(2)` or a large upper bound.
- Only first-order differential equations can be solved. Write the derivative as `diff(y, x)` .
- When x or another letter stays in the result, a form with e replaced by 2.718281828 may follow after "≈". To see only the exact form, tap AUTO at the top to switch to EXACT.
- Calculus works in graphs too: in "Graph" in the menu, write `x^3 − 3x` and `diff(x^3 − 3x)` on two lines to draw a function and its derivative together (4.8).

**Screen example** A limit typed with the Calculus keys (left) and a linear system solved with the Matrix keys (right; the preview shows the matrix as a table too).




## 4.5 Matrices and vectors

The Matrix page also has number keys, so a matrix can be typed without switching pages.

Write a matrix inside [ ], separate the elements with "," (the comma key in the row above) and the rows with ";".

- A 2 × 2 matrix: `[1, 2; 3, 4]`
- A column vector (3 × 1): `[1; 2; 3]`; a row vector: `[1, 2, 3]`
- Rows in their own brackets also work: `[[1, 2], [3, 4]]`

| Key | Writes | Meaning and example |
|---|---|---|
| [ ] ; | the character | Start and end of a matrix, and the end of a row |
| 0–9 and . | digits and the decimal point | The values of the elements |
| det | `det()` | Determinant. `det([1, 2; 3, 4])` = −2 |

| Key | Writes | Meaning and example |
|---|---|---|
| $A^T$ | `transpose()` | Transpose. `transpose([1, 2, 3; 4, 5, 6])` is a 3 × 2 matrix |
| $A^{-1}$ | `^(−1)` | Inverse. `[1, 2; 3, 4]^(−1)`. In an empty input field it inverts the last result (ans) |
| × − + | operators | Product, difference and sum of matrices. `[1, 2; 3, 4] × [5, 6; 7, 8]` |
| $I_n$ | `identity()` | Identity matrix. `identity(3)` is 3 × 3 |
| more… | — | Opens the list of functions at the category "Matrices & Vectors" (rank `rk`, row reduction `rref`, trace `trace`, adjugate `adj`, dot product `dot`, cross product `cross`, length `norm` and more; 51 in all) |

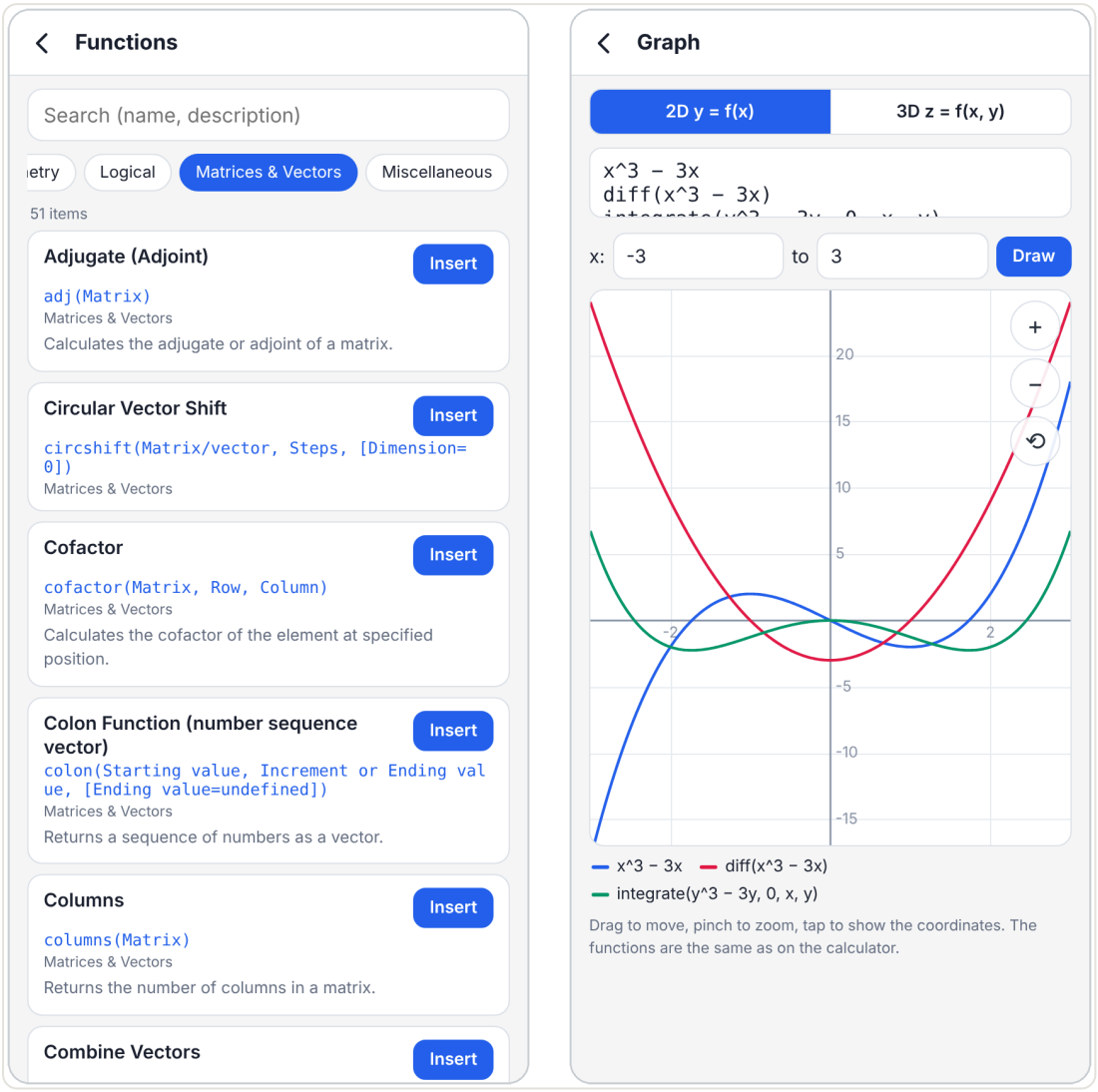
**How results are shown** A result with two or more rows is shown as a table in the history and in the preview. A wide matrix gets smaller characters; if it still does not fit, move it sideways. When a result is inserted into an expression or copied, it becomes one line of text such as `[−2  1; 1.5  −0.5]`, which works as an expression.

| To do this | Type | Result |
|---|---|---|
| Product of matrices | `[1, 2; 3, 4] × [5, 6; 7, 8]` | 2 × 2: 19, 22 / 43, 50 |
| Inverse | `[1, 2; 3, 4]^−1` | −2, 1 / 3/2, −1/2 (decimals shown beside it) |
| Determinant | `det([1, 2, 3; 4, 5, 6; 7, 8, 10])` | −3 |
| Linear system A x = b | `[2, 1; 1, 3]^−1 × [3; 5]` | the column vector 4/5, 7/5 (= 0.8, 1.4) |
| Row reduction | `rref([2, 1, 3; 1, 3, 5])` | 1, 0, 4/5 / 0, 1, 7/5 |
| Rank | `rk([1, 2; 2, 4])` | 1 |
| Trace | `trace([1, 2; 3, 4])` | 5 |
| Eigenvalues | `solve(det([2, 1; 1, 2] − x × identity(2)) = 0)` | [3 1] |
| Dot and cross product | `dot([1, 2, 3], [4, 5, 6])`, `cross([1, 0, 0], [0, 1, 0])` | 32, [0 0 1] |
| Length of a vector | `norm([3, 4])` | 5 |
| Sum and multiple | `[1, 2, 3] + 2 × [4, 5, 6]` | [9 12 15] |

### Good to know

- Two vectors cannot be multiplied with ×: "Please use the cross(), dot(), and hadamard() functions for vector multiplication" (cross product, dot product, element by element).
- The inverse of a matrix whose determinant is 0 shows "Inverse of singular matrix". A product of matrices whose sizes do not match says which sizes differ.
- Use x, y and z for letters in elements (a, b, c and the like are read as units).
- A matrix calculated with = becomes ans, so `det(ans)` or `ans × [1; 2]` go on from it.
- `trace()` was added by QCalc; Qalculate! does not have it. There is no function for eigenvalues; solve the characteristic equation det(A − xI) = 0 as in the table (it works for 2 × 2 and 3 × 3 matrices).

**Screen example** The list of functions opened with "more…" (left, the category "Matrices & Vectors") and a graph of x^3 − 3x with its derivative and integral (right).




## 4.6 Typing with the keyboard

Press ⌨ in the toolbar to type the expression with the phone's keyboard. The on-screen keys are hidden; the toolbar stays. Press ⌨ again, or close the keyboard, to get the keys back.

- Enter calculates (the same as =).
- The "=" inside an expression (an equation, or giving a variable a value) can be typed on the keyboard or with the a=b key. The big = key on the screen calculates.
- Write function and unit names in English (sin, sqrt, km, USD and so on). If you are not sure of a name, find it in the lists of the menu and tap "Insert". From the second letter typed, the functions, units and constants that begin with those letters are suggested below the input field (1.4.0). A tap writes the name; a function gets its parentheses with the cursor inside. From four letters, a word of the full name shown beside them finds them too ("pound" finds lb and GBP).
- A single * / ^ × or ÷ typed in the empty input field goes on from the last result (ans) (this can be turned off in the settings).

### Japanese input works too

When a conversion is confirmed, these characters are replaced automatically.

| Typed | Becomes | Example |
|---|---|---|
| Full-width digits, letters and signs | Half-width | １０＋５ → 10+5 |
| A full-width space | A half-width space | — |
| 円, ￥ | ¥ (Japanese yen) | 100円 to USD → 100¥ to USD |
| Hyphens such as ‐ - – — ― | − (minus) | — |
| "ー" between digits, parentheses or spaces | − (minus) | ５ー３ → 5−3 |

The phone's text size setting is not applied, so that the layout stays as designed. To make text larger, choose "Large" or "Extra large" under "Text size" in the app's settings (4.9).

## 4.7 Examples of expressions

Expressions are written the same way as in Qalculate!. The results below were calculated by this version of the engine (with the default settings). Expressions with "=" are typed with the a=b key or the keyboard (⌨). "Examples and tips" in the menu has 69 examples in 12 groups; tap one to put it into the input field (the screen example after the table). There are more calculus examples in 4.4 and matrix examples in 4.5.

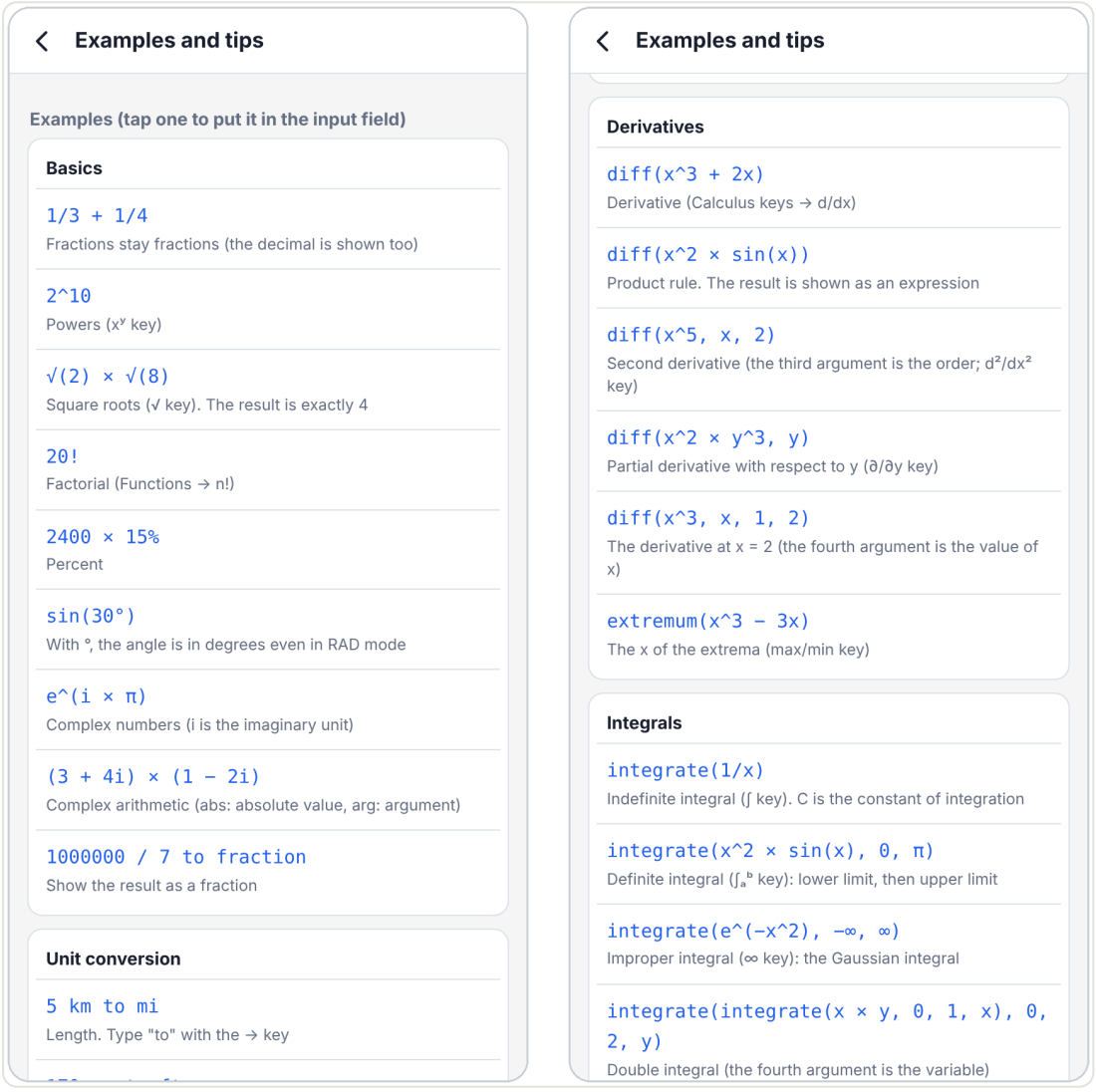
| To do this | Type | Result on the screen |
|---|---|---|
| Arithmetic (multiplication first) | 1+2×3 | = 7 |
| Fractions stay fractions | 1/3 + 1/4 | = 7/12 ≈ 0.5833333333 |
| Decimals without rounding errors | 0.1 + 0.2 | = 0.3 |
| Square roots (the exact form too) | √(8) | = 2 × √(2) ≈ 2.828427125 |
| Trigonometry in degrees | sin(30°) | = 1/2 = 0.5 |
| Large numbers | 2^100 | ≈ 1.267650600E30 (with "Exponential notation" set to "Never": = 1267650600228229401496703205376) |
| Factorial | 100! | ≈ 9.332621544E157 (all 158 digits with "Never" and a precision of 200) |
| Unit conversion | 5 km to mi | ≈ 3 mi + 188 yd + 2.393700787 in (≈ 3.106855961 mi with "Mixed units" off) |
| Calculations with units | 60 km/h × 2.5 h | = 150 km |
| Temperature | 20 °C to °F | = 68 °F |

| To do this | Type | Result on the screen |
|---|---|---|
| Currency | 100 USD to EUR | ≈ €… (with the current exchange rates) |
| Equation (solving = 0) | solve(x^2 − 5x + 6) | = [3  2] |
| Equation written with = | x^2 = 4 | x = 2 or x = −2 |
| Integral | integrate(x^2 sin(x), 0, pi) | = $\pi^2 - 4$ ≈ 5.869604401 |
| Derivative | diff(x^3) | = $3x^2$ |
| Sum | sum(n^2, 1, 10) | = 385 |
| Inverse matrix | [[1,2],[3,4]]^−1 | = [−2  1; 3/2  −1/2] = [−2  1; 1.5  −0.5] |
| Determinant | det([1, 2; 3, 4]) | = −2 |
| Limit | limit(sin(x)/x, 0) | = 1 (with the note on limits, 4.4) |
| Improper integral | integrate(e^(−x^2), −∞, ∞) | = √(π) ≈ 1.772453851 |
| Differential equation | dsolve(diff(y, x) = x^2) | = $x^3 / 3 + C$ |
| Traditional Japanese units | 30 tsubo to $m^2$ | ≈ 99.17355372 $m^2$ |
| Number bases | 255 to hex | = 0xFF |
| Input in other bases | 0b1010 + 0x1F | = 41 |
| Prime factors | factor 360 | = $2^3 \times 3^2 \times 5$ |
| Greatest common divisor | gcd(12, 18) | = 6 |
| Percentages | 2000 × 10% | = 200 |
| Remainder | 10 mod 3 | = 1 |
| Complex numbers | (1 + 2i)^2 | = −3 + 4i |
| Dates | today + 100 days | the date 100 days later (2027-02-17 when calculated on 2026-11-09) |
| Uncertainty propagation | (5 +/- 0.2) m × (3 +/- 0.1) m | ≈ 15.00±0.78 $m^2$ |
| Data sizes | 1 GiB to MB | = 1073.741824 MB |
| Physical constants | speed_of_light to km/s | = 299792.458 km/s |
| A temporary variable | x = 5, then x × 2 | = 5 (with a warning), = 10 |
| The last result | ans + 1 | the last result plus 1 |

### Notes on writing expressions

- Single-letter names may be read as units (a is are, g is gram, m is metre and so on). Check the parsed expression above the input field. x, y and z are unknowns.
- A variable made with "x = 5" is temporary and disappears when the app closes. Save variables and functions you want to keep under "User-defined variables and functions" in the menu.
- After "to" (the → key) you can write units and currencies, and also hex, bin, oct, fraction, factors and more.

**Screen example** "Examples and tips" in the menu (left) and its examples of derivatives and integrals (right). Tap an example to put it into the input field; the preview shows its result. "Undo" brings back the expression you were typing. The list ends with 12 tips.

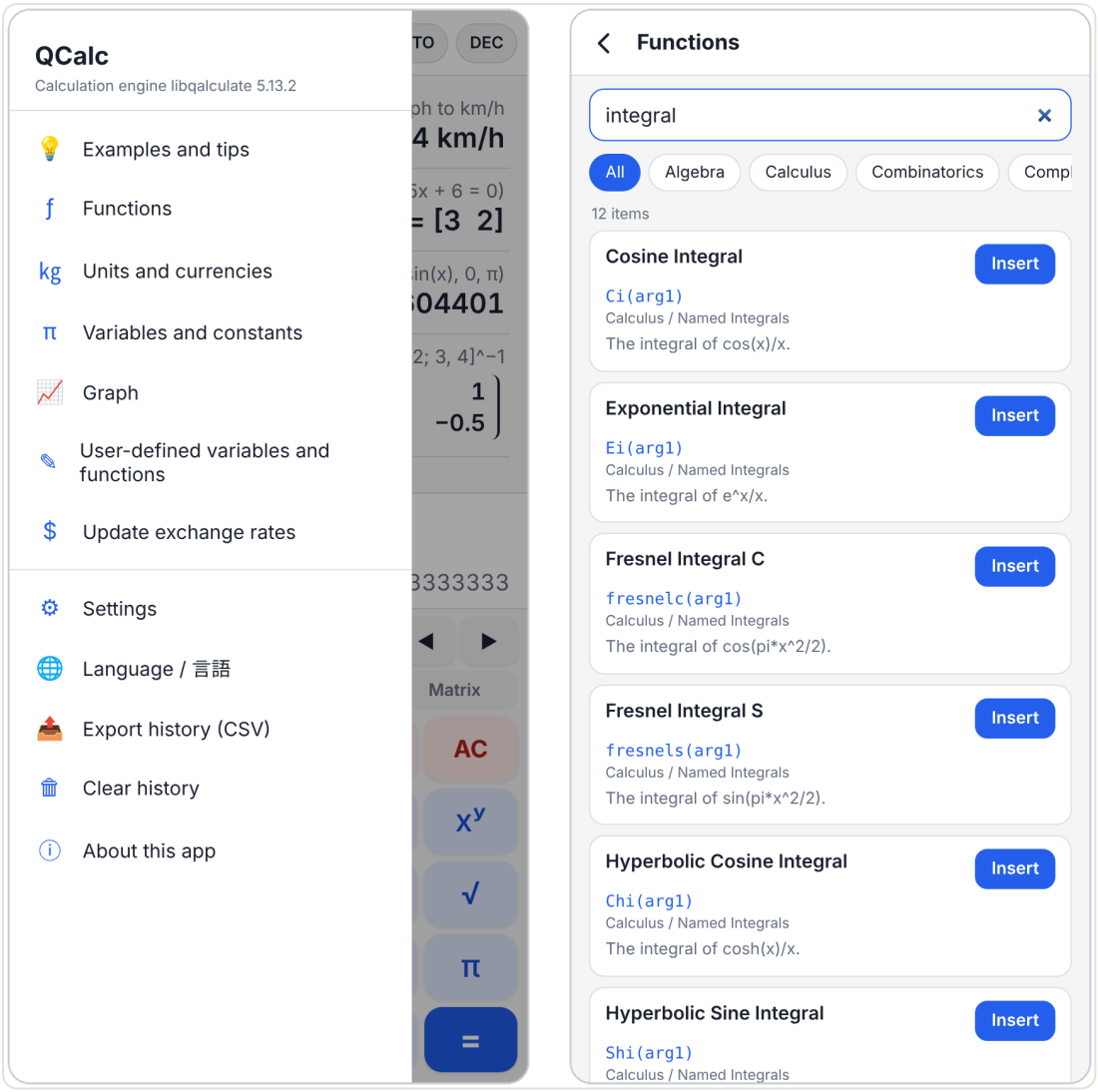



## 4.8 The menu

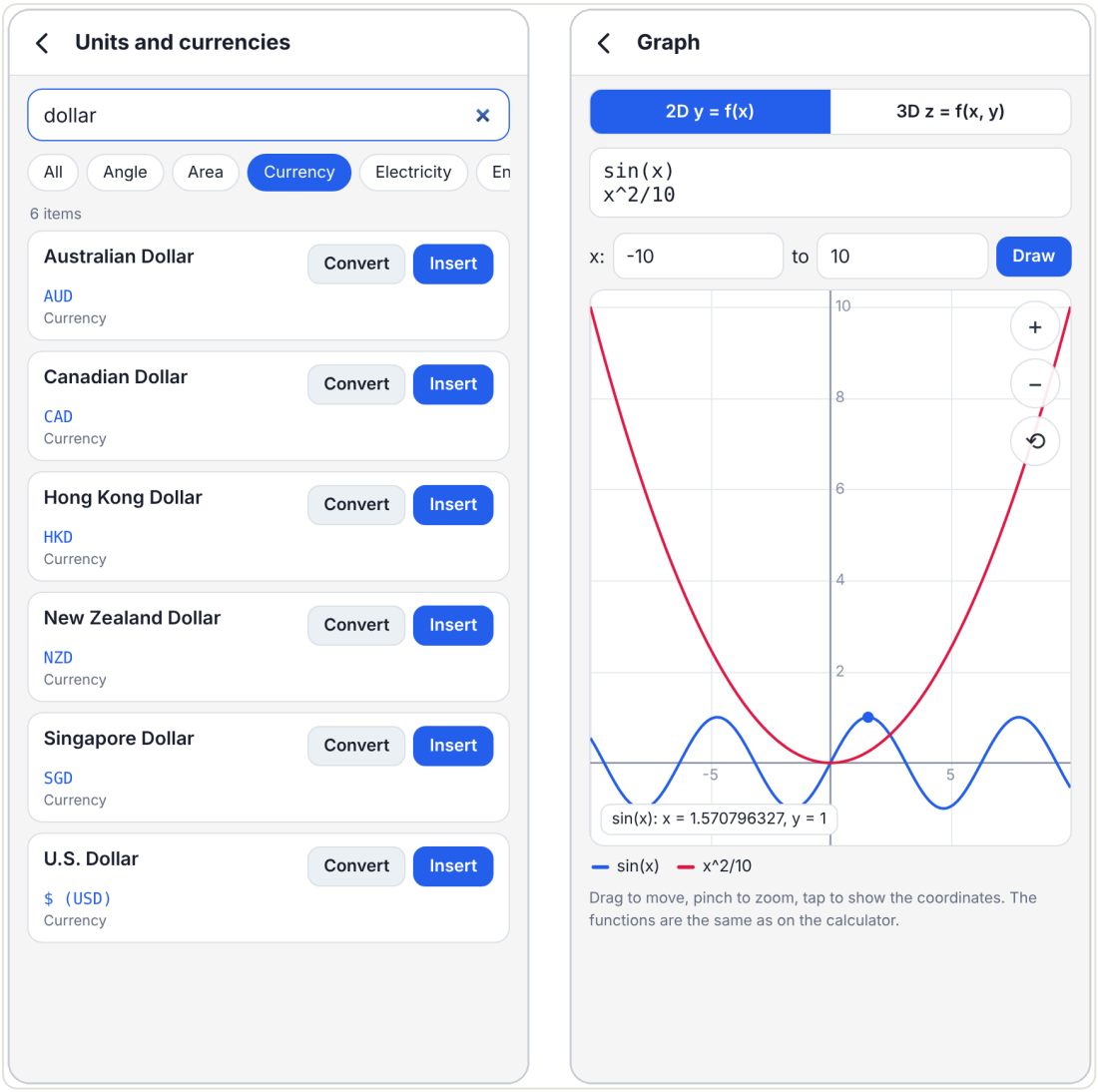
Open it with ☰ at the top left. Each panel closes with ‹ at its top left or with the back gesture.

| Menu item | What it does |
|---|---|
| Examples and tips | 69 examples in 12 groups (basics, units, currencies, equations, derivatives, integrals, limits, series and differential equations, matrices, vectors, number bases and integers, dates, statistics, uncertainty and constants) and 12 tips. Tap an example to put it into the input field (screen example in 4.7) |
| Functions | The functions by category. Search by name or description (on the Japanese screen, also in Japanese); the best match comes first. Tap an item to read its description. Opened with "more…" on the Calculus or Matrix page, it starts at that category. "Insert" writes "name()" with the cursor inside the parentheses. plot, command, load and export, which do not work in this app, are not listed |
| Units and currencies | Starts with the currencies. Search by name (on the Japanese screen, also in Japanese, such as 円 or メートル). The category "Japanese Units" has 10 traditional Japanese units: tsubo, tan, sun, shaku, ken, ri, gō, shō, monme and kan (QCalc adds them; Qalculate! does not have them. Example: 30 tsubo to $m^2$). "Insert" writes the unit's name, "Convert" writes " to " and its name |
| Variables and constants | π, e, physical constants and more, with their values. "Insert" writes the name |
| Graph | 2D graphs of y = f(x) and 3D graphs of surfaces z = f(x, y), switched with the buttons at the top. See "Graphs (2D and 3D)" at the end of this section. The expressions, the ranges and the last kind used are kept |
| User-defined variables and functions | Saves variables (a name and a value, such as r = 5 cm) and functions (arguments x, y, z; save hyp as sqrt(x^2 + y^2), then hyp(3, 4) = 5). They are kept for the next time. Saving with the same name changes one; "Delete" removes it. A variable made on the calculator with "x = 5" is shown as temporary and disappears when the app closes. Built-in names (sin, km and so on) cannot be used |
| Update exchange rates | Shows when the rates were last updated. "Update now" downloads them from three sources such as the European Central Bank (needs the internet). With the setting on, rates older than 7 days are updated when the app starts |
| Settings | 20 settings for language, calculation, display and input (4.9) |
| Language / 言語 | English or Japanese. The app opens again in the chosen language; the history and the expression being typed stay |
| Export history (CSV) | Saves the history as a CSV file where you choose (Android's save screen, for example in Downloads or Google Drive). The columns are date and time, expression, result as shown, and the value as a plain number when the result is one. The file is UTF-8 with a byte order mark, so Excel and other spreadsheets read it correctly |
| Clear history | After a confirmation, clears the whole history (ans and user definitions stay) |
| About this app | The versions of the app and of the engine, the phone and Android, the examples, privacy (with a link to the privacy policy), licenses ("Open source licenses": the bundled software and their full license texts), and links to QCalc's source code (GitHub) and the Qalculate! website |

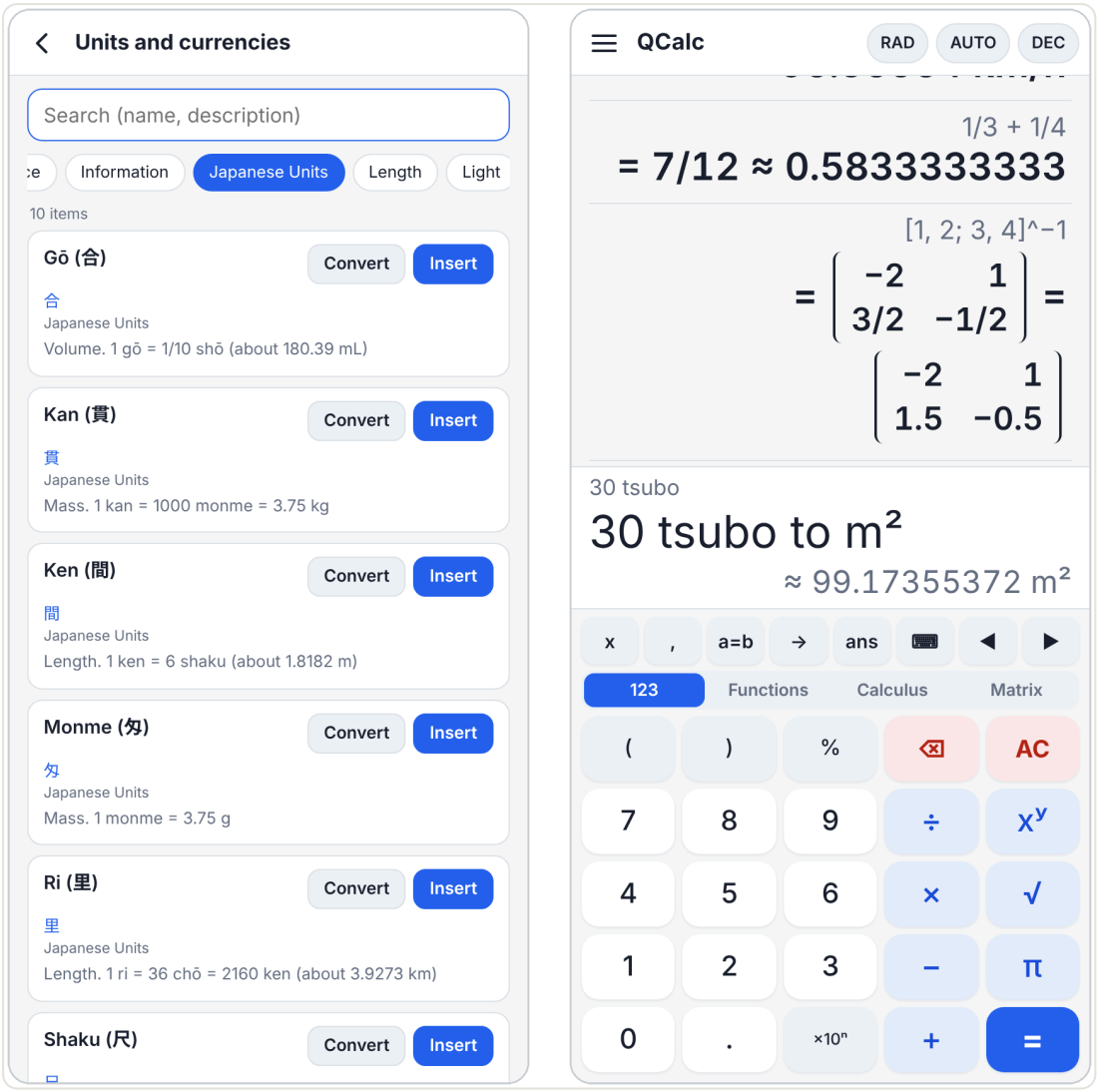
**Screen example** The menu (left) and the list of functions searched for "integral" (right: each function's name, title, syntax and category).



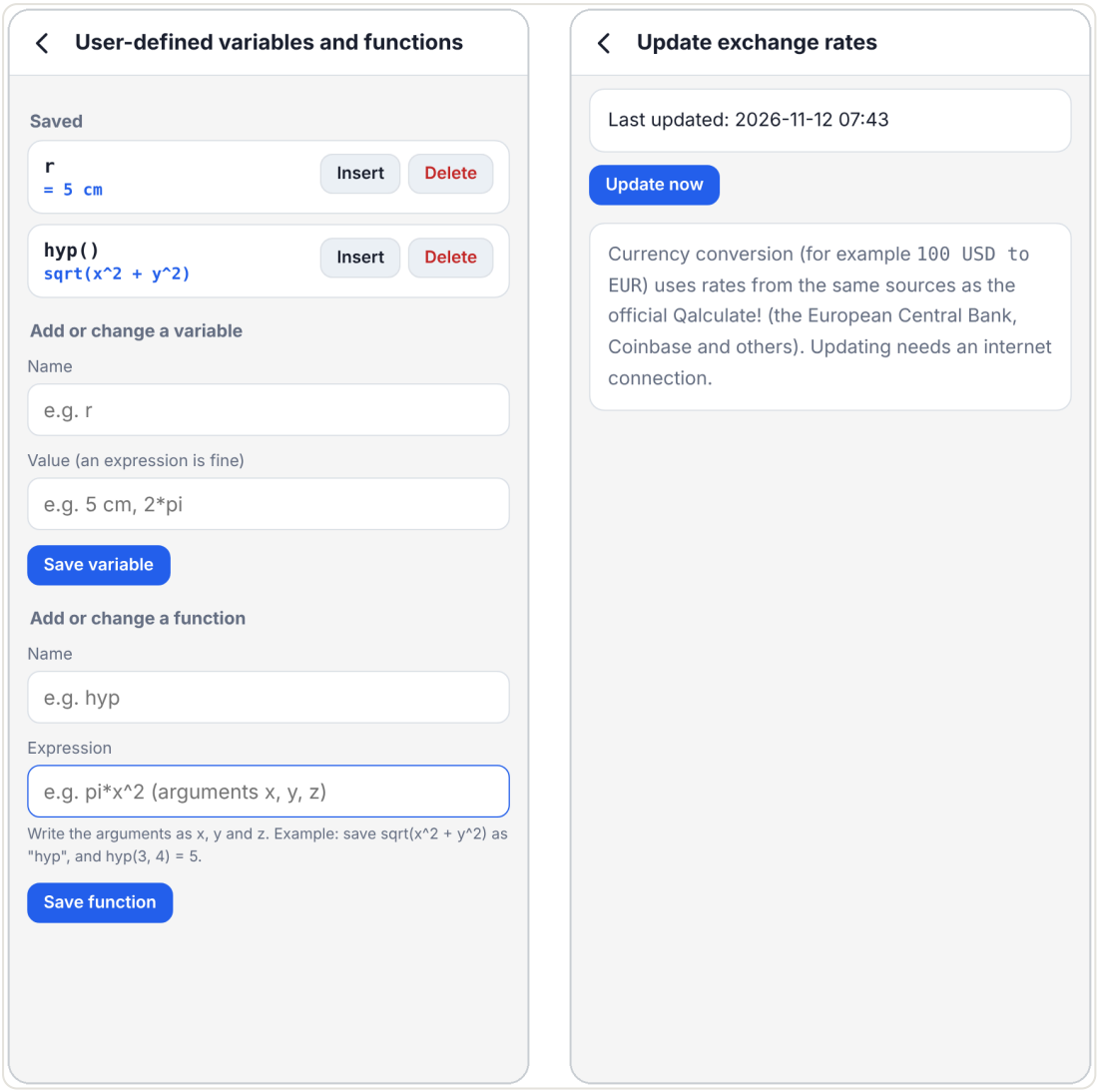
**Screen example** The units and currencies searched for "dollar" (left) and a graph of sin(x) and x^2/10 with a tapped point (right).



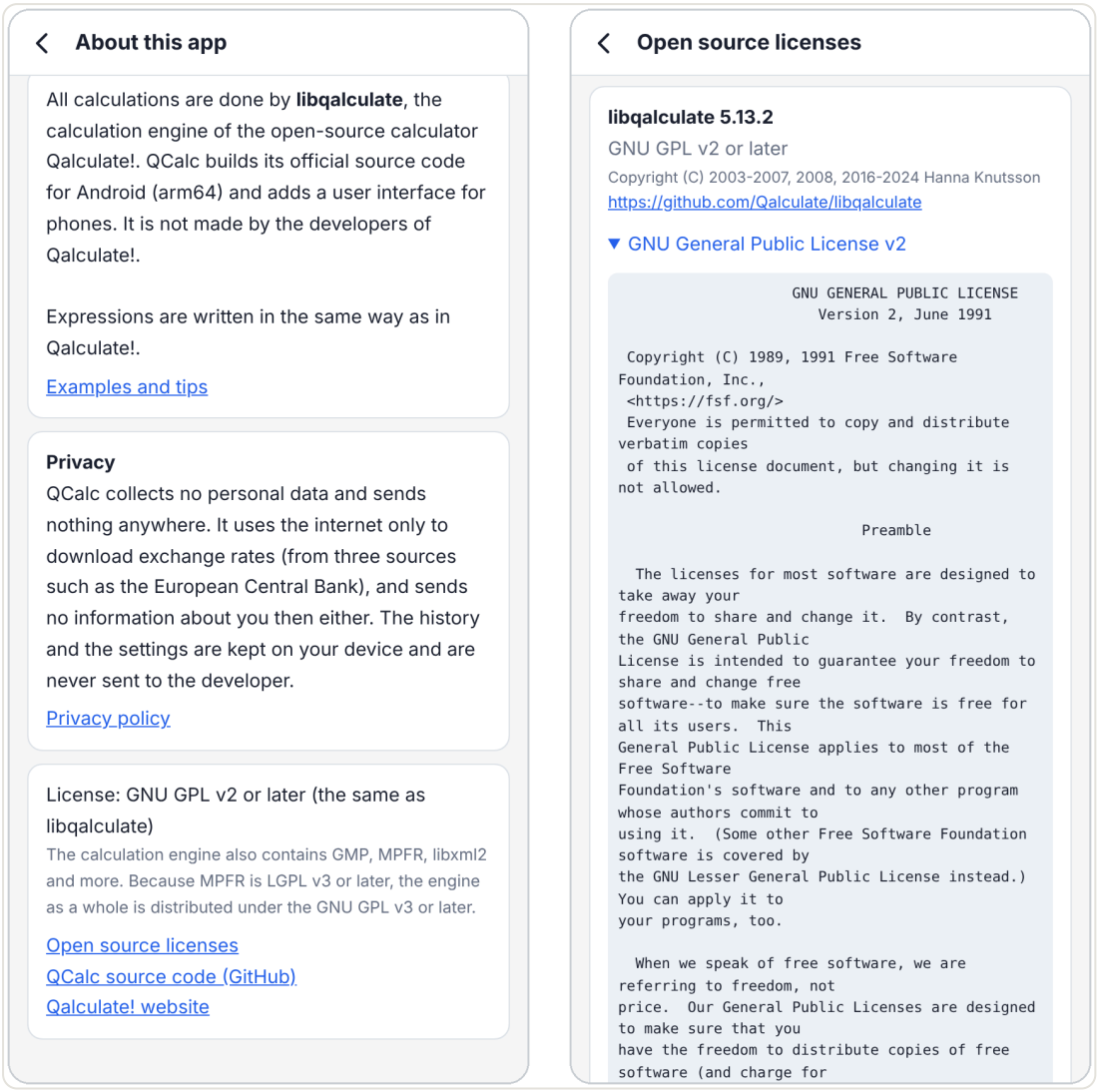
**Screen example** The category "Japanese Units" (left) and the calculator with the text size "Extra large" (right, converting 30 tsubo to square metres).



**Screen example** User definitions with the variable r (5 cm) and the function hyp (left) and the exchange-rate update (right).



**Screen example** The lower part of "About this app" (left: privacy and licenses) and the full license text of libqalculate in "Open source licenses" (right).




## Graphs (2D and 3D)

Open "Graph" in the menu and choose the kind at the top: "2D y = f(x)" or "3D z = f(x, y)".

### 2D graphs (y = f(x))

1. Write one function per line (variable x, up to 6), for example sin(x) and x^2/10. Derivatives such as diff(x^3 − 3x) and integrals with x as the upper bound, such as integrate(y^3 − 3y, 0, x, y), are drawn too (screen example in 4.5).
2. Enter the range of x and press "Draw" (expressions such as pi work too).
3. Drag to move, pinch to zoom. When the view moves, the missing part is calculated again automatically.
4. Tap near a line to show the coordinates of that point. ＋ and − zoom; ⟲ goes back to the range you entered.

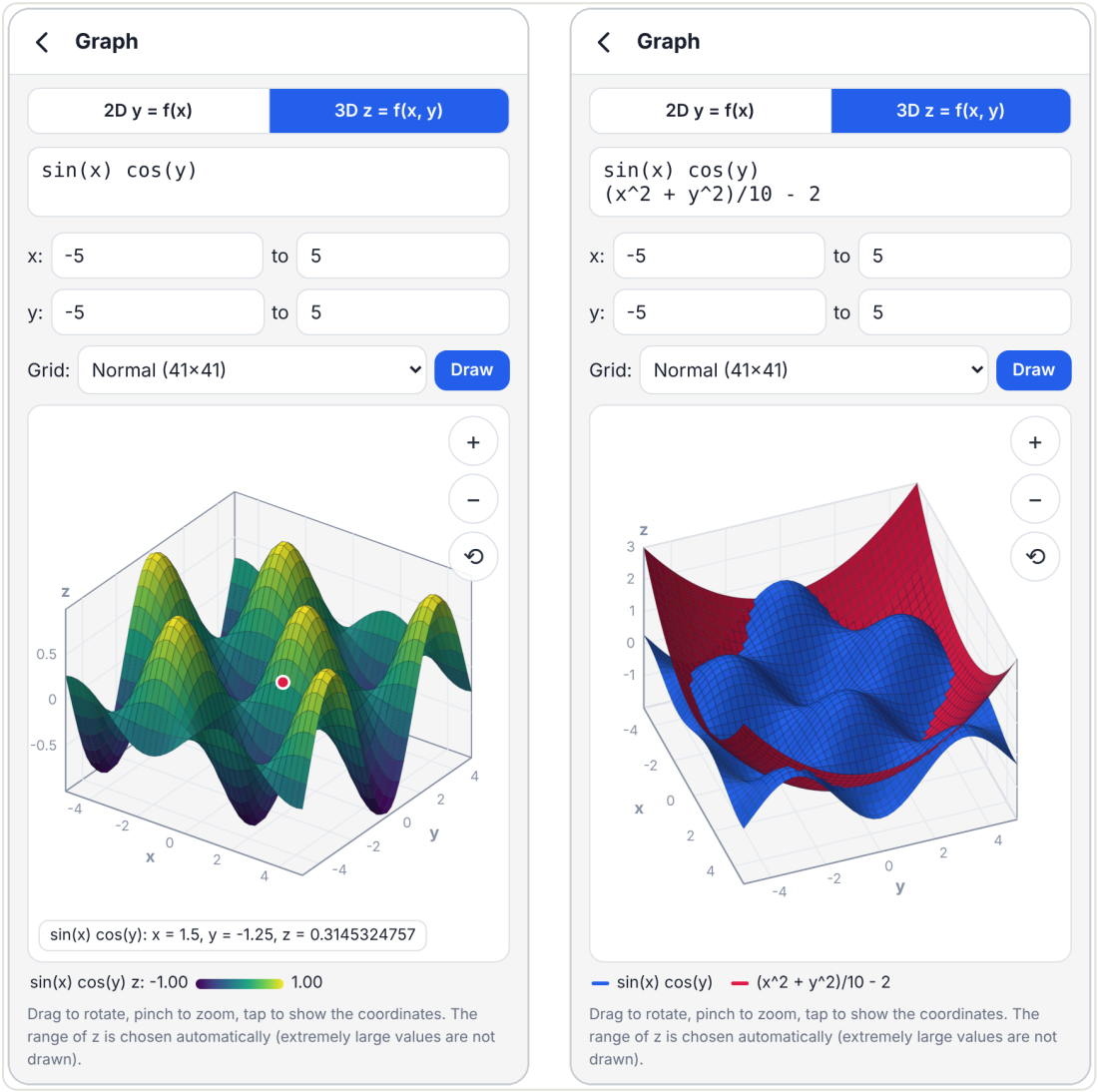
### 3D graphs (z = f(x, y))

1. Write one function per line (variables x and y, up to 4), for example sin(x) cos(y).

2. Enter the ranges of x and y, choose the grid (Coarse 25×25, Normal 41×41, Fine 61×61) and press "Draw".
3. Drag to turn, pinch to zoom. ＋ and － zoom; ⟲ goes back to the first view.
4. Tap the surface to show the coordinates (x, y, z) of the nearest grid point.

- One function is colored by height; two or more get one color each. With one function, the bar below shows which color is which z.
- Points where the value is not a real number (sqrt(x) for x < 0 and the like) and extremely large values are not drawn.
- The fine grid and many functions take longer. After 15 seconds, what was calculated so far is drawn.

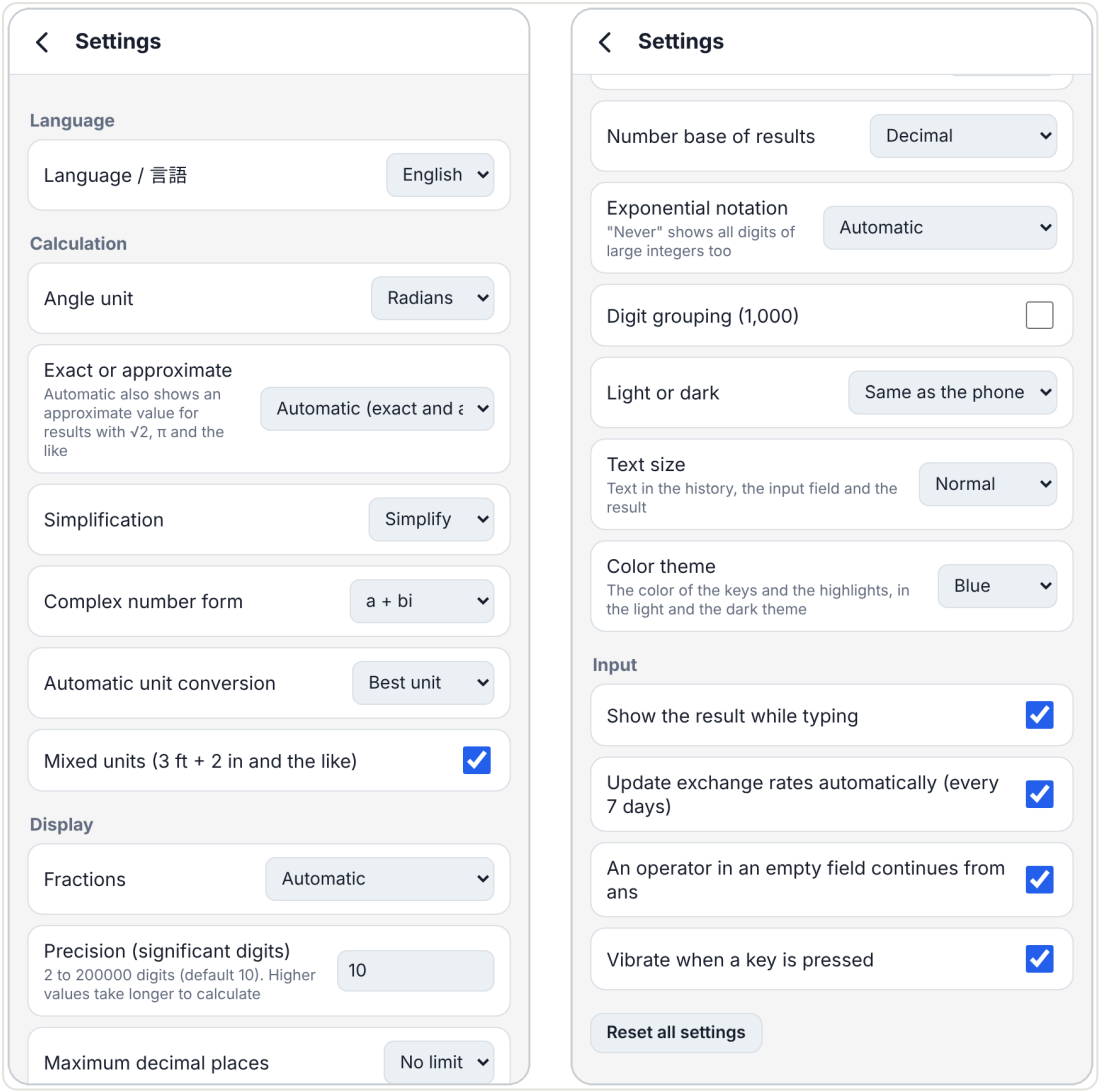
**Screen example** The surface "sin(x) cos(y)" with a tapped point (left) and two functions together, turned a little (right).




## 4.9 Settings

A setting is saved as soon as it is changed and applies at once, also to the expression being typed. "Reset all settings" at the bottom sets everything back to the defaults (the language stays).

**Screen example** The settings (left: the top, right: the bottom). The table below explains each item.




| Group | Setting | Values | Default | Description |
|---|---|---|---|---|
| Language | Language / 言語 | English, 日本語 | English | The language of the app (also in the menu) |
| Calculation | Angle unit | Radians, Degrees, Gradians | Radians | The angle of trigonometric functions. Also switched with a chip at the top |
| Calculation | Exact or approximate | Automatic (exact and approximate), Always exact, Always approximate (decimals) | Automatic | Automatic also shows an approximate value for results with √2, π and the like |
| Calculation | Simplification | Simplify, Factorize, Expand | Simplify | The form of symbolic results |
| Calculation | Complex number form | a + bi, Exponential, Polar, cis | a + bi | How complex results are written |
| Calculation | Automatic unit conversion | Best unit, Best SI unit, Base units, None | Best unit | Which unit a result with units is shown in |

| Group | Setting | Values | Default | Description |
|---|---|---|---|---|
| Calculation | Mixed units (3 ft + 2 in and the like) | On, Off | On | Off shows decimals of a single unit |
| Display | Fractions | Automatic, Decimal, Decimal (when exact), Fraction, Mixed fraction | Automatic | Automatic shows the fraction and the decimal (7/12 ≈ 0.5833333333) |
| Display | Precision (significant digits) | 2 to 200000 | 10 | The precision of calculations. Higher values take longer ("Very long results" in 4.3) |
| Display | Maximum decimal places | No limit, 0, 2, 4, 6, 8, 10 | No limit | Rounds decimals to this many places |
| Display | Number base of results | Decimal, Hexadecimal, Binary, Octal, Duodecimal, Roman numerals | Decimal | The chip switches between decimal, hexadecimal, binary and octal |
| Display | Exponential notation | Automatic, Scientific, Engineering (exponent a multiple of 3), Always, Never (all digits) | Automatic | "Never" shows all digits of large integers too |
| Display | Digit grouping (1,000) | On, Off | Off | Groups of three digits before the decimal point, with commas (12,345,678); the decimals get narrow spaces (3.141 592 654) |
| Display | Light or dark | Same as the phone, Light, Dark | Same as the phone | The light or the dark colors of the screen. "Same as the phone" follows the phone's dark theme. Light and Dark keep that look whatever the phone's setting is (1.4.0) |
| Display | Text size | Normal, Large, Extra large | Normal | Text in the history, the input field and the result. Large is 1.18 times Normal, Extra large 1.38 times. The keys and the menu do not change |
| Display | Color theme | Blue, Green, Teal, Purple, Rose, Orange, Graphite | Blue | The color of the keys and highlights, in the light and the dark theme |
| Input | Show the result while typing | On, Off | On | The preview |
| Input | Update exchange rates automatically (every 7 days) | On, Off | On | Updates when the app starts and the rates are older than 7 days (tried at most every 6 hours) |
| Input | An operator in an empty field continues from ans | On, Off | On | ÷ × + ^ and the like in an empty input field put "ans" in front |

| Group | Setting | Values | Default | Description |
|---|---|---|---|---|
| Input | Vibrate when a key is pressed | On, Off | On | A short vibration (8 milliseconds) |

## 4.10 Troubleshooting

| Problem | Cause and what to do |
|---|---|
| QCalc is not found in Google Play or cannot be installed | On a phone that is not 64-bit (arm64) or has Android older than 7.0, it is not shown and cannot be installed. Until the app is published, it can be installed only from the test link with a Google account registered as a tester |
| The app shows Japanese (or English) and you want the other language | Menu → "Language / 言語" (4.8) |
| "QCalc needs Android System WebView, which is not available…" | Update Android System WebView in the Play Store, or turn it on in the Settings app, then open QCalc again |
| "Could not start the calculation engine" | Tap the message to try again. If that does not help, close and open the app; if it still fails, reinstall it |
| The preview stays at "…" | The preview stops after 1.5 seconds. Press = to calculate for up to 20 seconds |
| "The calculation took too long and was stopped" | Make the expression simpler, lower the precision, or choose "Always approximate" |
| "Too long to copy" | Very long results, such as a complex number with 200,000-digit parts, cannot be copied because of a limit in Android (about 1 MB at once). Calculate again with a lower precision |
| "The middle of this long result is not shown (N digits)" | Results longer than about 450,000 characters are shown as their beginning, "…" and their end. Copying the result copies the same. Lower the precision, or use ans in the next calculation (ans keeps all the digits) |
| An old result in the history ends with "…" | The history keeps the newest results in full up to 1,000,000 characters in all; older long results keep their first 2,000 characters. Tap the expression and calculate again to get all the digits |
| Tapping a long result does not insert it | Results longer than 20,000 characters are not inserted. Use the ans key (the last result) |
| "The calculation engine stopped unexpectedly. Try a simpler expression." | This happens with expressions that need too much memory. The engine starts again with the next calculation and ans is restored (ans2 to ans5 and temporary variables are lost) |
| Large numbers look like 1.267650600E30 | Set "Exponential notation" to "Never". For values with decimals, raise the precision too |
| 5 km to mi gives "3 mi + 188 yd + …" | Turn off "Mixed units" in the settings |

| Problem | Cause and what to do |
|---|---|
| A single-letter name is calculated as a unit (a × 2 gets the unit a and the like) | a (are), g (gram), m (metre) and others are unit names. Check the parsed expression. After "a = 3", the name is a variable until the app closes (with a warning). Use other names for variables you keep |
| Currency conversion is out of date, or "It has been N days since the exchange rates last were updated" | Menu → "Update exchange rates" → "Update now" (needs the internet) |
| The exchange-rate update fails | Check the connection. If only some sources failed, their host names are shown, and the rates from the other sources were updated |
| "This result is too long to insert (use the ans key)" | Very long results are kept shortened in the history. The last result works with the ans key |
| "The saved user definitions could not be read, so their file was moved aside" | The file of definitions was damaged. Save the variables and functions you need again |
| A graph is not drawn | "Check the range of x" or "Check the range of y": make sure the left value is smaller than the right one. 'Press "Draw" to draw the graph': the last drawing stopped part way; press "Draw". In 3D, "There are no real values (check the ranges of x and y)": the values in that range are complex or the like (sqrt(x) for x < 0) |
| The back gesture closes the app | When no menu or panel is open, the app moves to the background. The expression and the history are kept |
| "The screen stopped again and again. Close the app and open it again." | Close and open the app. If it happens again, update Android System WebView |
| You want the default settings back | "Reset all settings" at the bottom of the settings |
| A 3D graph takes long or is drawn only in part | Choose the grid "Coarse" or "Normal", or fewer functions. After 15 seconds, what was calculated so far is drawn |
| Part of a 3D graph is missing | Extremely large values (near 0 in 1/(x y) and the like) are left out so that the surface does not flatten. A smaller range shows more detail |
| You want to read the licenses or the privacy policy | Menu → "About this app". The privacy policy and the source code open in the browser |
| You cleared an expression or the history by mistake | Tap "Undo", which appears right after: for an expression until you type something, for a history item for 5 seconds, for the whole history for 6 seconds (4.3) |
| The text is too small | Set "Text size" to "Large" or "Extra large" (4.9) |
| The "=" of an equation cannot be typed (= calculates) | Use the a=b key in the row above the keys. The big = key calculates |
| "limit() is incomplete and unreliable" | A note shown with every limit; it is not a fault. Check important results another way, by putting in nearby values or drawing a graph (4.4) |
| An integral or a differential equation comes back unchanged, or "Unable to integrate the expression" | The engine could not find a closed form. For a definite integral, use numbers as bounds or romberg for numerical integration; for an infinite sum, use a large upper bound (4.4) |

| Problem | Cause and what to do |
| --- | --- |
| Matrix calculations show "Inverse of singular matrix", "Please use the cross(), dot(), and hadamard() functions…" or "The second matrix must have as many rows…" | The inverse of a matrix whose determinant is 0, × between two vectors, or a product whose sizes do not match (4.5). Check the row separator ";" and the element separator "," too |
| You want to know how to use QCalc and what you can write | Menu → "Examples and tips". Tap an example to try it. Functions, units and constants can be searched in the lists of the menu |
| The app stays light when the phone is in the dark theme (or the other way round) | The setting "Light or dark" is Light or Dark. Set it to "Same as the phone" and the app follows the phone again (4.9) |